



**Department of Electrical, Computer
& Biomedical Engineering**
Faculty of Engineering & Architectural Science

Course Title:	Fundamentals of Data Engineering
Course Number:	COE848
Semester/Year (e.g. F2016)	W2026

Instructor:	Dr. Yasaman Ahmadiadli
Lab Report Number	Lab 6
Lab Title	Lab 6: Semi-Structured Data

Due Date	April 2, 2026
Submission Date	April 2, 2026

Last Name	First Name	Student Number	Section	Signature*
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*By signing above you attest that you have contributed to this written lab report and confirm that all work you have contributed to this lab report is your own work. Any suspicion of copying or plagiarism in this work will result in an investigation of Academic Misconduct and may result in a "0" on the work, an "F" in the course, or possibly more severe penalties, as well as a Disciplinary Notice on your academic record under the Student Code of Academic Conduct, which can be found online at: <http://www.ryerson.ca/senate/current/pol60.pdf>

5 XSLT Queries:

1) I would like to see the details and capacity for all the Restaurants

```
<?xml version="1.0"?>
<xsl:stylesheet version="1.0"
xmlns:xsl="http://www.w3.org/1999/XSL/Transform">

<xsl:template match="/">
<html>
<body>
<h2>All Restaurants</h2>

<xsl:for-each select="RestaurantSystem/Restaurants/Restaurant">
  <p>
    ID: <xsl:value-of select="restaurant_id"/><br/>
    Name: <xsl:value-of select="name"/><br/>
    Cuisine: <xsl:value-of select="cuisine_type"/><br/>
    Capacity: <xsl:value-of select="total_capacity"/>
  </p>
  <hr/>
</xsl:for-each>

</body>
</html>
</xsl:template>

</xsl:stylesheet>
```

All Restaurants

ID: 1
Name: Villa Grill
Cuisine: Mediterranean
Capacity: 80

ID: 2
Name: Sunset Sushi
Cuisine: Japanese
Capacity: 60

ID: 3
Name: Pasta Palace
Cuisine: Italian
Capacity: 70

ID: 4
Name: Spice Route
Cuisine: Indian
Capacity: 75

ID: 5
Name: Taco Town
Cuisine: Mexican
Capacity: 50

ID: 6
Name: Dragon Wok
Cuisine: Chinese
Capacity: 65

ID: 7
Name: Burger Barn
Cuisine: American
Capacity: 90

ID: 8
Name: Green Garden
Cuisine: Vegetarian
Capacity: 55

ID: 9
Name: Seaside Grill
Cuisine: Seafood
Capacity: 85

ID: 10
Name: Golden Curry
Cuisine: Thai
Capacity: 60

2) I would like to see the details and Registration Information from all the current Customers

```
<?xml version="1.0"?>
<xsl:stylesheet version="1.0"
xmlns:xsl="http://www.w3.org/1999/XSL/Transform">

<xsl:template match="/">
<html>
<body>
<h2>Customers</h2>

<xsl:for-each select="RestaurantSystem/Customers/Customer">
  <p>
    Name: <xsl:value-of select="first_name"/>
    <xsl:value-of select="last_name"/><br/>
    Email: <xsl:value-of select="email"/>
  </p>
</xsl:for-each>

</body>
</html>
</xsl:template>

</xsl:stylesheet>
```

Customers

Name: BobSmith
Email: bob@email.com

Name: AliceBrown
Email: alice@email.com

Name: JohnLee
Email: john@email.com

Name: MariaGarcia
Email: maria@email.com

Name: DavidKim
Email: david@email.com

Name: SaraWilson
Email: sara@email.com

Name: OmarHassan
Email: omar@email.com

Name: LinaPatel
Email: lina@email.com

Name: ChrisJohnson
Email: chris@email.com

Name: EmmaTaylor
Email: emma@email.com

Name: JAVA_INSERTION_TESTJAAV_INSERTION_TEST
Email: java@email.com

Name: TESTTEST
Email: TEST@email.com

3) I would like to see and specifically filter by the restaurants with Total Capacities larger than 80 people, typically the larger restaurants

```
<?xml version="1.0"?>
<xsl:stylesheet version="1.0"
xmlns:xsl="http://www.w3.org/1999/XSL/Transform">

<xsl:template match="/">
<html>
<body>
<h2>Large Restaurants (Capacity > 80)</h2>

<xsl:for-each select="RestaurantSystem/Restaurants/Restaurant[total_capacity > 80]">
  <p>
    <xsl:value-of select="name"/> - Capacity:
    <xsl:value-of select="total_capacity"/>
  </p>
</xsl:for-each>

</body>
</html>
</xsl:template>

</xsl:stylesheet>
```

Large Restaurants (Capacity > 80)

Burger Barn - Capacity: 90

Seaside Grill - Capacity: 85

4) I would like to see and specifically filter the available catalog of menu items for all restaurants to find items that are less than \$10:

```
<?xml version="1.0"?>
<xsl:stylesheet version="1.0"
xmlns:xsl="http://www.w3.org/1999/XSL/Transform">

<xsl:template match="/">
<html>
<body>
<h2>Menu Items Under $10</h2>

<xsl:for-each select="RestaurantSystem/MenuItems/MenuItem[price < 10]">
  <p>
    Item: <xsl:value-of select="item_name"/><br/>
    Price: <xsl:value-of select="price"/>
  </p>
</xsl:for-each>

</body>
</html>
</xsl:template>

</xsl:stylesheet>
```

Menu Items Under \$10

Item: Mango Juice
Price: 5.0

Item: Chicken Taco
Price: 9.0

5) I would like to see ALL the reservations that have been made and are processed into the system:

```
<?xml version="1.0"?>
<xsl:stylesheet version="1.0"
xmlns:xsl="http://www.w3.org/1999/XSL/Transform">

<xsl:template match="/">
<html>
<body>
<h2>Reservations</h2>

<xsl:for-each select="RestaurantSystem/Reservations/Reservation">
  <p>
    Reservation ID: <xsl:value-of select="reservation_id"/><br/>
    Customer ID: <xsl:value-of select="customer_id"/><br/>
    Date: <xsl:value-of select="reservation_date"/><br/>
    Time: <xsl:value-of select="reservation_time"/>
  </p>
  <hr/>
</xsl:for-each>

</body>
</html>
</xsl:template>

</xsl:stylesheet>
```

Reservations

Reservation ID: 1
Customer ID: 1
Date: 2026-04-09
Time: 18:00

Reservation ID: 2
Customer ID: 2
Date: 2026-04-10
Time: 19:00

Reservation ID: 3
Customer ID: 3
Date: 2026-04-11
Time: 18:30

Reservation ID: 4
Customer ID: 4
Date: 2026-04-11
Time: 20:00

Reservation ID: 5
Customer ID: 5
Date: 2026-04-12
Time: 17:00

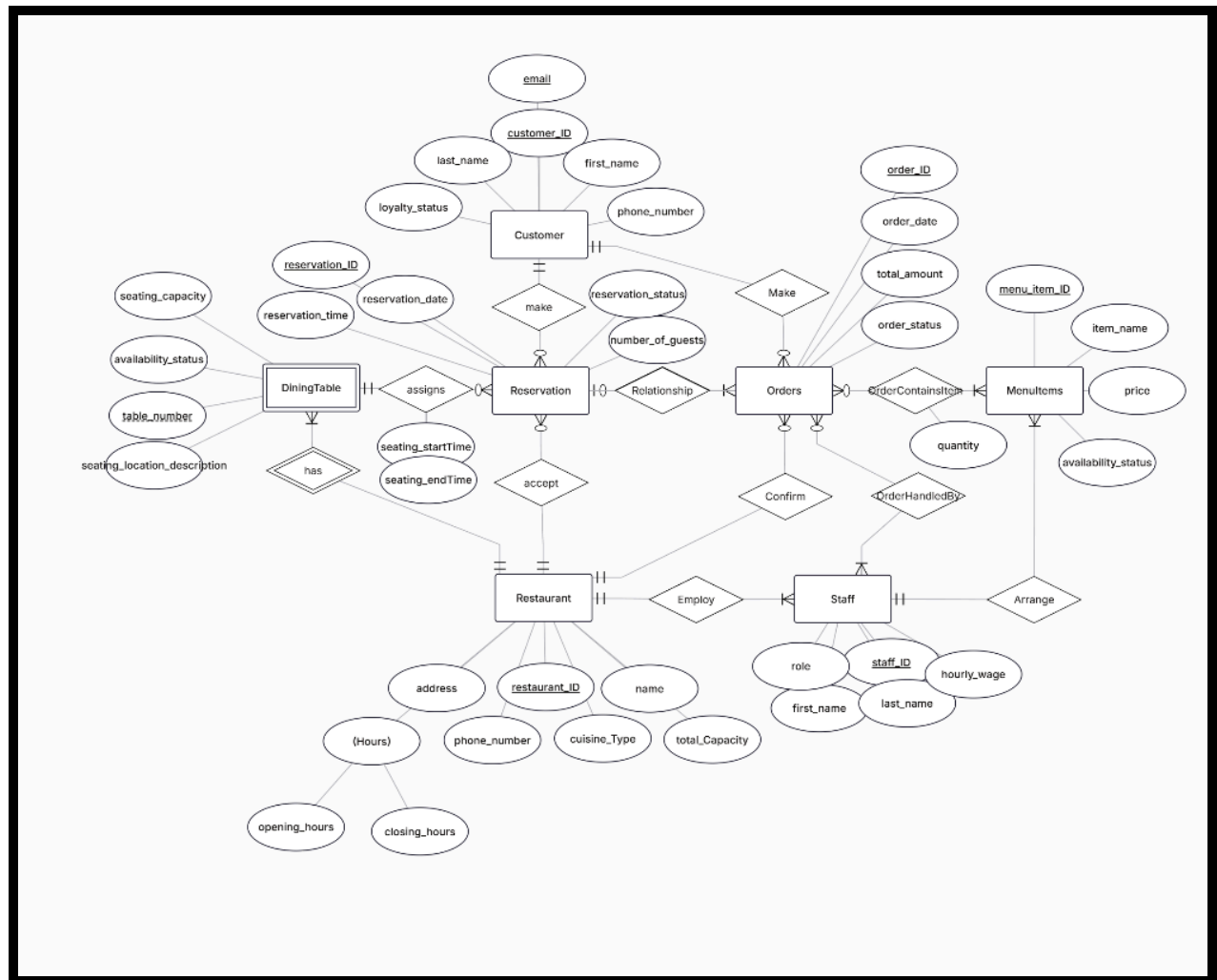
Reservation ID: 6
Customer ID: 6
Date: 2026-04-12
Time: 19:30

Reservation ID: 7
Customer ID: 7
Date: 2026-04-13
Time: 18:00

Reservation ID: 8
Customer ID: 8
Date: 2026-04-13
Time: 17:30

Reservation ID: 9
Customer ID: 9
Date: 2026-04-14
Time: 20:00

Reservation ID: 10
Customer ID: 10
Date: 2026-04-15
Time: 18:30



The ER diagram developed earlier from lab 2 and 3 was very much successfully translated into the XML schema in this lab, as we mapped each entity and relationship into appropriately structured XML elements. The main entities in the ER diagram that were the table names were Restaurant, Customer, Staff, DiningTable, MenuItem, Orders, and Reservations, all of which correspond to separate XML elements (this is basically tables) within the main root that is “RestaurantSystem” element. Within each element consists of sub-elements that identify each entity’s attributes (like restaurant name, phone number, price...).

Now the relationships from the ER diagram were properly preserved in this XML schema (.xsd). There were complicated relationships like OrderHandledBy and OrderContainsItem which derived from many-to-many relationships in the relational model; for many-to-many relationships, a dedicated and separate table is created, such that this requires another XML entity creation for those two relationships. These XML elements also had their Foreign Key incorporated (two FK since each from the two entities that made up the relationship). For the one-to-many relationship, the Foreign Key was still used and incorporated for references to its dependent entity.

Primary keys and foreign keys were also implemented from the relational schema, and was defined in the Schema Definition (.xsd) as “<xs:key>” and “<xs:keyref>”. The “DiningTable” entity also incorporated composite keys,

since it was a weak entity that is entirely dependent on the “Restaurant” entity, as such it used two entities with its primary key in the form of the two tags of, “<xs:field>”, specifying its own two primary key variables.

Now the XML method was slightly different from the traditional relational schema as it had different purposes. XML design required a much simpler representation of the schema design, much simpler than done in the ER diagram. XML prioritized keys and clean data storage over relationships and it involved different structuring methods. The XML grouped all its entities under the one single root element, which is the main file’s element. XML represents any relationship by simply writing the keys and these keys practically represent the data relationships.

Overall, the transformation from ER diagram to XML schema was certainly successful and it maintained the entire logical structure and key behaviours/functionalities of the database while adapting to its special hierarchical and data-storage oriented format which is what XML is known for.

restaurantSystem.xsd:

```
<?xml version="1.0"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">

  <xs:element name="RestaurantSystem">
    <xs:complexType>
      <xs:sequence>

        <!-- Restaurants -->
        <xs:element name="Restaurants">
          <xs:complexType>
            <xs:sequence>
              <xs:element name="Restaurant" maxOccurs="unbounded">
                <xs:complexType>
                  <xs:sequence>
                    <xs:element name="restaurant_id" type="xs:int"/>
                    <xs:element name="name" type="xs:string"/>
                    <xs:element name="address" type="xs:string"/>
                    <xs:element name="phone_number" type="xs:string"/>
                    <xs:element name="cuisine_type" type="xs:string"/>
                    <xs:element name="opening_hours" type="xs:string"/>
                    <xs:element name="closing_hours" type="xs:string"/>
                    <xs:element name="total_capacity" type="xs:int"/>
                  </xs:sequence>
                </xs:complexType>
              </xs:element>
            </xs:sequence>
          </xs:complexType>
        </xs:element>

        <!-- Customers -->
        <xs:element name="Customers">
          <xs:complexType>
            <xs:sequence>
              <xs:element name="Customer" maxOccurs="unbounded">
                <xs:complexType>
                  <xs:sequence>
                    <xs:element name="customer_id" type="xs:int"/>
                    <xs:element name="first_name" type="xs:string"/>
                    <xs:element name="last_name" type="xs:string"/>
                    <xs:element name="email" type="xs:string"/>
                    <xs:element name="phone_number" type="xs:string" minOccurs="0"/>
                    <xs:element name="loyalty_status" type="xs:string" minOccurs="0"/>
                  </xs:sequence>
                </xs:complexType>
              </xs:element>
            </xs:sequence>
          </xs:complexType>
        </xs:element>
      </xs:sequence>
    </xs:complexType>
  </xs:element>
</xs:schema>
```

```

        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
</xs:element>

<!-- Staff -->
<xs:element name="Staffs">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="Staff" maxOccurs="unbounded">
        <xs:complexType>
          <xs:sequence>
            <xs:element name="staff_id" type="xs:int"/>
            <xs:element name="restaurant_id" type="xs:int"/>
            <xs:element name="first_name" type="xs:string"/>
            <xs:element name="last_name" type="xs:string"/>
            <xs:element name="role" type="xs:string"/>
            <xs:element name="hourly_wage" type="xs:decimal"/>
          </xs:sequence>
        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
</xs:element>

<!-- Dining Tables -->
<xs:element name="DiningTables">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="DiningTable" maxOccurs="unbounded">
        <xs:complexType>
          <xs:sequence>
            <xs:element name="restaurant_id" type="xs:int"/>
            <xs:element name="table_number" type="xs:int"/>
            <xs:element name="seating_capacity" type="xs:int"/>
            <xs:element name="availability_status" type="xs:string"/>
            <xs:element name="seating_location_description" type="xs:string"/>
          </xs:sequence>
        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
</xs:element>

<!-- Menu Items -->

```

```
<xs:element name="MenuItems">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="MenuItem" maxOccurs="unbounded">
        <xs:complexType>
          <xs:sequence>
            <xs:element name="menu_item_id" type="xs:int"/>
            <xs:element name="staff_id" type="xs:int"/>
            <xs:element name="item_name" type="xs:string"/>
            <xs:element name="price" type="xs:decimal"/>
            <xs:element name="availability_status" type="xs:string"/>
          </xs:sequence>
        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
</xs:element>
```

```
<!-- Orders -->
<xs:element name="Orders">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="Order" maxOccurs="unbounded">
        <xs:complexType>
          <xs:sequence>
            <xs:element name="order_id" type="xs:int"/>
            <xs:element name="customer_id" type="xs:int"/>
            <xs:element name="restaurant_id" type="xs:int"/>
            <xs:element name="reservation_id" type="xs:int"/>
            <xs:element name="order_date" type="xs:string"/>
            <xs:element name="total_amount" type="xs:decimal"/>
            <xs:element name="order_status" type="xs:string"/>
          </xs:sequence>
        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
</xs:element>
```

```
<!-- Reservations -->
<xs:element name="Reservations">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="Reservation" maxOccurs="unbounded">
        <xs:complexType>
          <xs:sequence>
```

```

    <xs:element name="reservation_id" type="xs:int"/>
    <xs:element name="customer_id" type="xs:int"/>
    <xs:element name="restaurant_id" type="xs:int"/>
    <xs:element name="table_number" type="xs:int"/>
    <xs:element name="reservation_date" type="xs:string"/>
    <xs:element name="reservation_time" type="xs:string"/>
    <xs:element name="number_of_guests" type="xs:int"/>
    <xs:element name="reservation_status" type="xs:string"/>
  </xs:sequence>
</xs:complexType>
</xs:element>
</xs:sequence>
</xs:complexType>
</xs:element>

```

```

<!-- OrderHandledBy -->
<xs:element name="OrderHandledBy">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="OrderHandled" maxOccurs="unbounded">
        <xs:complexType>
          <xs:sequence>
            <xs:element name="order_id" type="xs:int"/>
            <xs:element name="staff_id" type="xs:int"/>
          </xs:sequence>
        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
</xs:element>

```

```

<!-- OrderContainsItems -->
<xs:element name="OrderContainsItems">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="OrderItem" maxOccurs="unbounded">
        <xs:complexType>
          <xs:sequence>
            <xs:element name="order_id" type="xs:int"/>
            <xs:element name="menu_item_id" type="xs:int"/>
            <xs:element name="quantity" type="xs:int"/>
          </xs:sequence>
        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>

```

```

    </xs:element>

</xs:sequence>
</xs:complexType>

<!-- PRIMARY KEYS -->
<xs:key name="restaurantKey">
  <xs:selector xpath="Restaurants/Restaurant"/>
  <xs:field xpath="restaurant_id"/>
</xs:key>

<xs:key name="customerKey">
  <xs:selector xpath="Customers/Customer"/>
  <xs:field xpath="customer_id"/>
</xs:key>

<xs:key name="staffKey">
  <xs:selector xpath="Staffs/Staff"/>
  <xs:field xpath="staff_id"/>
</xs:key>

<xs:key name="menuItemKey">
  <xs:selector xpath="MenuItems/MenuItem"/>
  <xs:field xpath="menu_item_id"/>
</xs:key>

<xs:key name="orderKey">
  <xs:selector xpath="Orders/Order"/>
  <xs:field xpath="order_id"/>
</xs:key>

<xs:key name="reservationKey">
  <xs:selector xpath="Reservations/Reservation"/>
  <xs:field xpath="reservation_id"/>
</xs:key>

<!-- COMPOSITE KEY (DiningTable) -->
<xs:key name="diningTableKey">
  <xs:selector xpath="DiningTables/DiningTable"/>
  <xs:field xpath="restaurant_id"/>
  <xs:field xpath="table_number"/>
</xs:key>

<!-- FOREIGN KEYS -->
<xs:keyref name="staffRestaurantFK" refer="restaurantKey">
  <xs:selector xpath="Staffs/Staff"/>

```



```

    <xs:field xpath="restaurant_id"/>
  </xs:keyref>

  <xs:keyref name="menuStaffFK" refer="staffKey">
    <xs:selector xpath="MenuItems/MenuItem"/>
    <xs:field xpath="staff_id"/>
  </xs:keyref>

  <xs:keyref name="orderCustomerFK" refer="customerKey">
    <xs:selector xpath="Orders/Order"/>
    <xs:field xpath="customer_id"/>
  </xs:keyref>

  <xs:keyref name="orderRestaurantFK" refer="restaurantKey">
    <xs:selector xpath="Orders/Order"/>
    <xs:field xpath="restaurant_id"/>
  </xs:keyref>

  <xs:keyref name="orderReservationFK" refer="reservationKey">
    <xs:selector xpath="Orders/Order"/>
    <xs:field xpath="reservation_id"/>
  </xs:keyref>

  <xs:keyref name="reservationCustomerFK" refer="customerKey">
    <xs:selector xpath="Reservations/Reservation"/>
    <xs:field xpath="customer_id"/>
  </xs:keyref>

  <xs:keyref name="reservationRestaurantFK" refer="restaurantKey">
    <xs:selector xpath="Reservations/Reservation"/>
    <xs:field xpath="restaurant_id"/>
  </xs:keyref>

  <!-- COMPOSITE FK -->
  <xs:keyref name="reservationTableFK" refer="diningTableKey">
    <xs:selector xpath="Reservations/Reservation"/>
    <xs:field xpath="restaurant_id"/>
    <xs:field xpath="table_number"/>
  </xs:keyref>

  <xs:keyref name="orderHandledOrderFK" refer="orderKey">
    <xs:selector xpath="OrderHandledBy/OrderHandled"/>
    <xs:field xpath="order_id"/>
  </xs:keyref>

  <xs:keyref name="orderHandledStaffFK" refer="staffKey">

```

```

    <xs:selector xpath="OrderHandledBy/OrderHandled"/>
    <xs:field xpath="staff_id"/>
  </xs:keyref>

  <xs:keyref name="orderItemOrderFK" refer="orderKey">
    <xs:selector xpath="OrderContainsItems/OrderItem"/>
    <xs:field xpath="order_id"/>
  </xs:keyref>

  <xs:keyref name="orderItemMenuFK" refer="menuItemKey">
    <xs:selector xpath="OrderContainsItems/OrderItem"/>
    <xs:field xpath="menu_item_id"/>
  </xs:keyref>

</xs:element>

</xs:schema>

```

restaurantSystem.xml:

```

<?xml version="1.0"?>
<RestaurantSystem>

  <Restaurants>
    <Restaurant><restaurant_id>1</restaurant_id><name>Villa Grill</name><address>123 King
    St
    Toronto</address><phone_number>4165551000</phone_number><cuisine_type>Mediterranea
    n</cuisine_type><opening_hours>10:00</opening_hours><closing_hours>22:00</closing_hou
    rs><total_capacity>80</total_capacity></Restaurant>
    <Restaurant><restaurant_id>2</restaurant_id><name>Sunset Sushi</name><address>22
    Queen St
    Toronto</address><phone_number>4165551001</phone_number><cuisine_type>Japanese</cu
    isine_type><opening_hours>11:00</opening_hours><closing_hours>23:00</closing_hours><t
    otal_capacity>60</total_capacity></Restaurant>
    <Restaurant><restaurant_id>3</restaurant_id><name>Pasta Palace</name><address>77
    Dundas St
    Toronto</address><phone_number>4165551002</phone_number><cuisine_type>Italian</cuisi
    ne_type><opening_hours>10:00</opening_hours><closing_hours>22:00</closing_hours><tota
    l_capacity>70</total_capacity></Restaurant>
    <Restaurant><restaurant_id>4</restaurant_id><name>Spice Route</name><address>90 Bloor
    St
    Toronto</address><phone_number>4165551003</phone_number><cuisine_type>Indian</cuisi
    ne_type><opening_hours>11:00</opening_hours><closing_hours>23:00</closing_hours><tota
    l_capacity>75</total_capacity></Restaurant>
    <Restaurant><restaurant_id>5</restaurant_id><name>Taco Town</name><address>15 Front

```

St

Toronto</address><phone_number>4165551004</phone_number><cuisine_type>Mexican</cuisine_type><opening_hours>10:00</opening_hours><closing_hours>21:00</closing_hours><total_capacity>50</total_capacity></Restaurant>

<Restaurant><restaurant_id>6</restaurant_id><name>Dragon Wok</name><address>55 Yonge St

Toronto</address><phone_number>4165551005</phone_number><cuisine_type>Chinese</cuisine_type><opening_hours>11:00</opening_hours><closing_hours>22:30</closing_hours><total_capacity>65</total_capacity></Restaurant>

<Restaurant><restaurant_id>7</restaurant_id><name>Burger Barn</name><address>300 College St

Toronto</address><phone_number>4165551006</phone_number><cuisine_type>American</cuisine_type><opening_hours>10:00</opening_hours><closing_hours>22:00</closing_hours><total_capacity>90</total_capacity></Restaurant>

<Restaurant><restaurant_id>8</restaurant_id><name>Green Garden</name><address>88 Bay St

Toronto</address><phone_number>4165551007</phone_number><cuisine_type>Vegetarian</cuisine_type><opening_hours>09:00</opening_hours><closing_hours>21:00</closing_hours><total_capacity>55</total_capacity></Restaurant>

<Restaurant><restaurant_id>9</restaurant_id><name>Seaside Grill</name><address>12 Harbour St

Toronto</address><phone_number>4165551008</phone_number><cuisine_type>Seafood</cuisine_type><opening_hours>11:00</opening_hours><closing_hours>23:00</closing_hours><total_capacity>85</total_capacity></Restaurant>

<Restaurant><restaurant_id>10</restaurant_id><name>Golden Curry</name><address>9 Spadina Ave

Toronto</address><phone_number>4165551009</phone_number><cuisine_type>Thai</cuisine_type><opening_hours>11:00</opening_hours><closing_hours>22:00</closing_hours><total_capacity>60</total_capacity></Restaurant>

</Restaurants>

<Customers>

<Customer><customer_id>1</customer_id><first_name>Bob</first_name><last_name>Smith</last_name><email>bob@email.com</email><phone_number>4161110001</phone_number><loyalty_status>Gold</loyalty_status></Customer>

<Customer><customer_id>2</customer_id><first_name>Alice</first_name><last_name>Brown</last_name><email>alice@email.com</email><phone_number>4161110002</phone_number><loyalty_status>Silver</loyalty_status></Customer>

<Customer><customer_id>3</customer_id><first_name>John</first_name><last_name>Lee</last_name><email>john@email.com</email><phone_number>4161110003</phone_number><loyalty_status>None</loyalty_status></Customer>

<Customer><customer_id>4</customer_id><first_name>Maria</first_name><last_name>Garcia</last_name><email>maria@email.com</email><phone_number>4161110004</phone_number><loyalty_status>Gold</loyalty_status></Customer>

<Customer><customer_id>5</customer_id><first_name>David</first_name><last_name>Kim</last_name><email>david@email.com</email><phone_number>4161110005</phone_number>

```
r><loyalty_status>Silver</loyalty_status></Customer>
<Customer><customer_id>6</customer_id><first_name>Sara</first_name><last_name>Wilson</last_name><email>sara@email.com</email><phone_number>4161110006</phone_number><loyalty_status>None</loyalty_status></Customer>
<Customer><customer_id>7</customer_id><first_name>Omar</first_name><last_name>Hassan</last_name><email>omar@email.com</email><phone_number>4161110007</phone_number><loyalty_status>Gold</loyalty_status></Customer>
<Customer><customer_id>8</customer_id><first_name>Lina</first_name><last_name>Patel</last_name><email>lina@email.com</email><phone_number>4161110008</phone_number><loyalty_status>Silver</loyalty_status></Customer>
<Customer><customer_id>9</customer_id><first_name>Chris</first_name><last_name>Johnson</last_name><email>chris@email.com</email><phone_number>4161110009</phone_number><loyalty_status>None</loyalty_status></Customer>
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